The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Foundation-Generating Substrate Root Dynamics
Necessity / Uniqueness from
Root-Generating Substrate Seed Dynamics

Abstract

The current benchmark-facing frontier on the active primitive-reservoir observer-reduction line is no longer the ground-generating substrate foundation package $\mathfrak{F}^{\mathrm{sub,fdn}}$. The foundation-forcing theorem already removed that stopping point by proving that $\mathfrak{F}^{\mathrm{sub,fdn}}$ is a canonical and unique output of the deeper foundation-generating substrate root package $\mathfrak{R}^{\mathrm{sub,rt}}$. The open burden therefore moved below foundation scope to root scope.

The present note takes the primary route specified by the current routing layer. It introduces a still more primitive explicit substrate class, *root-generating substrate seed dynamics* $\mathfrak{S}^{\mathrm{sub,seed}}$, and proves that the recorded root package is itself a canonical and unique projected exact-shell output of that seed layer. For each sector the theorem isolates the benchmark-relevant seed core: the seed-to-root transport, the seed fiber-minimizer ladder over the fixed root, foundation, ground, origin, precursor, lift, and substrate fibers, the exact seed shell/support/zero transport, the charted zero/hidden split, the hidden charge, the exact seed zero-response realization, and the fixed-data solvability / coercive package. It then proves that every seed-faithful presentation inducing the fixed root package determines the same root output up to canonical seed-faithful equivalence.

In that precise sense the foundation-generating substrate root dynamics are no longer merely available inside a deeper class. They are necessary and unique at benchmark-facing scope as the projected exact-shell output of root-generating substrate seed dynamics. The benchmark boundary remains fixed throughout: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. The honest burden therefore moves below root scope to the seed layer itself.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has now reached the point where another same-output root relay would be benchmark neutral. The current foundation-forcing theorem already removed the ground-generating substrate foundation package $\mathfrak{F}^{\mathrm{sub,fdn}}$ as the live stopping point on the active primitive-reservoir observer-reduction line. Once that theorem is on record, the next honest benchmark-facing move must go one level lower and address the foundation-generating substrate root dynamics that support foundation scope.

There are two disciplined ways to do that. One can derive the root package from still more primitive explicit substrate structure, or one can prove a genuinely smaller theorem-level collapse strictly inside the current root package. The present note takes the first route. Its gain is not a completed first-principles derivation of gravity. Its narrower claim is that the recorded root

package is no longer an independently inserted stopping point on the active one-metric line: it is the canonical projected exact-shell output of a deeper seed package.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not alter the benchmark exact-closure package. It does not introduce a second operative metric or a new low-energy matter law. It only proves that the current root layer is itself forced once a deeper seed package is fixed.

2. Fixed Primitive Continuation Data

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Remark 1. *Definition 1 is not reopened here. The primitive continuation data, the one operative metric, and the ordinary matter route remain fixed. The present note only addresses the remaining benchmark-facing freedom at root scope.*

3. Recorded Root Target

Definition 2 (Recorded foundation-generating substrate root package)**.** *Fix, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, the recorded root package*

$$\mathfrak{R}_{\sigma}^{\mathrm{sub,rt}}.$$

At benchmark-facing scope this package consists of the following data.

1. *A root state space and compact convex root body over the recorded foundation, ground, origin, precursor, lift, and substrate bodies, together with affine projections to those lower bodies and unique root-to-foundation, root-to-ground, root-to-origin, root-to-precursor, root-to-lift, and root-to-substrate fiber minimizers.*

2. *A root restriction section whose zero set on the root body is the exact root shell.*

3. *Affine root observer-data and shell-response readouts factoring through the recorded foundation readouts already used by the current foundation-forcing theorem.*

4. *A root shell chart to the same reservoir body, a root shell-internal support recorder, a root support shell, a root support zero core, support-shell and zero charts, and the same charted zero/hidden split in the same equation variables.*

5. *A root hidden charge together with the corresponding hidden recorder, an observer-compatible exact root zero-response realization, and the same fixed-data solvability and coercive control at benchmark scope.*

Remark 2. *Definition 2 is the exact target already fixed by the current root-facing frontier. The present note does not enlarge that target. It asks whether any benchmark-relevant freedom remains in the deeper seed layer once that exact root package is required.*

4. Root-Generating Substrate Seed Dynamics

Definition 3 (Root-generating substrate seed dynamics)**.** *For each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a* root-generating substrate seed dynamics package

$$\mathfrak{S}_{\sigma}^{\mathrm{sub,seed}}$$

consists of the following data.

1. *A compact convex seed body over the recorded root state space together with affine projection to root scope and unique seed-to-root, seed-to-foundation, seed-to-ground, seed-to-origin, seed-to-precursor, seed-to-lift, and seed-to-substrate fiber minimizers.*

2. *A seed restriction section whose zero set on the seed body is the exact seed shell and whose projected exact-shell transport recovers the recorded root restriction section and exact root shell.*

3. *Seed observer-data and shell-response readouts, a seed shell chart to the same reservoir body, a seed shell-internal support recorder, a seed support shell, a seed support zero core, support-shell*

and zero charts, and a charted zero/hidden split whose projected exact-shell transport recovers the corresponding root readouts, shell/support transport, zero core, and zero/hidden split.

4. *A seed hidden charge, an observer-compatible exact seed zero-response realization, and a fixed-data solvability / coercive package whose projected exact-shell transport recovers the corresponding root hidden charge, exact root zero-response realization, solvability statement, and coercive control.*

The package is benchmark faithful *if it preserves the benchmark outputs* (1)*, introduces no second operative metric, introduces no enlarged low-energy matter law, and is presented as a deeper substrate-side source for the current root package rather than as a replacement benchmark theory.*

Definition 4 (Seed-faithful presentation)**.** *A sectorwise seed presentation is* seed faithful *if its projected exact-shell transport reproduces exactly the recorded root package of Definition 2: the same root-to-foundation, root-to-ground, root-to-origin, root-to-precursor, root-to-lift, and root-to-substrate fiber minimizer images, the same exact root shell, the same shell chart to the same reservoir body, the same support shell and support zero core, the same charted zero/hidden split in the same equation variables, the same hidden charge and exact root zero-response realization, and the same fixed-data solvability / coercive package. It must also leave the fixed benchmark outputs* (1) *unchanged.*

Definition 5 (Benchmark-relevant seed core)**.** *For a seed-faithful presentation, the* benchmark-relevant seed core

$$\mathfrak{S}_{\sigma}^{\mathrm{sub,seed,core}}$$

is the part of the seed package consisting of:

1. *the seed-to-root transport together with the seed root-, foundation-, ground-, origin-, precursor-, lift-, and substrate-fiber minimizer images;*
2. *the exact seed shell, seed support shell, and seed support zero core;*
3. *the shell/support transport and the charted zero/hidden split that project to the recorded root shell/support transport;*
4. *the seed hidden charge, the exact seed zero-response realization, and the fixed-data solvability / coercive package.*

Definition 6 (Seed-faithful equivalence)**.** *A* seed-faithful equivalence *between two seed-faithful presentations is an identification of their cores* $\mathfrak{S}_{\sigma}^{\mathrm{sub,seed,core}}$ *that preserves the same projected root package, the same minimizer ladder, the same shell/support transport, the same zero/hidden coordinates, the same hidden charge, the same exact seed zero-response realization, and the same solvability / coercive package.*

Remark 3. *Definition 6 is intentionally narrower than full off-shell identity of the ambient seed body. The active burden is not to classify every possible seed thickening outside the benchmark-relevant seed core. It is to remove the benchmark-relevant freedom still visible at root scope on the active one-metric line.*

5. Necessity of the Root Package from Seed Dynamics

Proposition 1 (The fiber-minimizer ladder is necessary)**.** *Every seed-faithful presentation determines the same root-, foundation-, ground-, origin-, precursor-, lift-, and substrate-fiber minimizer images*

inside the seed layer. Hence the seed-to-root transport relevant at benchmark scope is canonical.

Proof. By Definition 4, the seed presentation must recover exactly the fixed root package of Definition 2. In particular, it must recover the same root body and the same lower bodies together with the same unique minimizers on the corresponding recorded fibers. Since those lower fibers are fixed continuation data, the image of each seed minimizer is keyed uniquely by the lower point it lifts. The benchmark-relevant seed minimizer ladder therefore depends only on the shared lower datum and is canonical. □

Proposition 2 (Exact-shell transport data are necessary)**.** *Every seed-faithful presentation determines the same exact-shell transport data at benchmark scope. More precisely, the seed-to-root projection restricts to diffeomorphic transport from the seed shell, support shell, and zero core onto the corresponding fixed root loci, and the root shell chart, support recorder, support-shell chart, zero chart, and zero/hidden split are thereby forced up to canonical seed-faithful equivalence.*

Proof. If a seed presentation is faithful, then by definition its projected exact-shell transport recovers precisely the shell-side data already fixed in Definition 2. The seed shell must therefore project diffeomorphically to the recorded root shell, otherwise the fixed root shell would not be recovered as exact-shell image. The same argument applies one step further to the support shell and then to the zero core. Once those projected loci are fixed, the root shell chart to the common reservoir body, the support recorder, the support-shell chart, the zero chart, and the zero/hidden split cannot vary independently: changing any of them would change the induced root shell/support transport or the fixed equation-side coordinates already recorded on the active line. Hence all these constituents are forced up to canonical seed-faithful equivalence. □

Proposition 3 (Hidden charge, zero-response realization, solvability, and coercivity are necessary)**.** *Every seed-faithful presentation determines, up to seed-faithful equivalence, the same seed hidden charge data, the same exact seed zero-response realization, the same fixed-data solvability statement, and the same coercive control on data-invisible directions.*

Proof. The fixed root package already records one hidden sector, one hidden charge with positive gap, one exact root zero-response realization, and one solvability / coercive package on the root-side zero core. A seed-faithful presentation must project to that same root package. Because the shell/support transport and the zero/hidden coordinates are already fixed by Proposition 2, the hidden sector cannot be changed without changing the induced root recorder. Likewise, the exact seed zero-response realization must project to the same visible, current, and equilibrium shell data, so its image on the seed zero core is forced by the same root compatibility requirement. The solvability and coercive statements are properties of this same projected zero-response core; if they differed at seed scope, the induced root statements would differ as well. Therefore those constituents are necessary up to seed-faithful equivalence. □

Proposition 4 (The benchmark package remains fixed)**.** *Every seed-faithful presentation preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly* $g^\sharp$*, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of Definition 4. The present note removes only seed-level freedom internal to the active derivational line. It does not alter the recorded primitive outputs. □

6. Main Theorem

Theorem 1 (Foundation-generating substrate root dynamics necessity / uniqueness from root-generating substrate seed dynamics)**.** *Fix the primitive continuation data of Definition 1 and the recorded root package of Definition 2. Then, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, the benchmark-relevant seed core* $\mathfrak{S}_\sigma^{\mathrm{sub,seed,core}}$ *underlying a seed-faithful presentation is necessary and unique at benchmark scope up to seed-faithful equivalence. Equivalently, the recorded root package* $\mathfrak{R}_\sigma^{\mathrm{sub,rt}}$ *is a canonical and unique projected exact-shell output of root-generating substrate seed dynamics. More precisely:*

1. *every seed-faithful presentation must determine the same root-, foundation-, ground-, origin-, precursor-, lift-, and substrate-fiber minimizer ladder and the same exact-shell transport data described in Definition 5;*

2. *for any two seed-faithful presentations, there is a unique seed-faithful equivalence between their benchmark-relevant cores;*

3. *under that equivalence, the projected root shell, support shell, zero core, shell/support/equation transport, hidden charge, exact seed zero-response realization, fixed-data solvability statement, and coercive control all coincide at benchmark scope;*

4. *consequently no additional benchmark-relevant root-level freedom remains on the active one-metric line beyond seed-faithful relabeling of* $\mathfrak{S}_\sigma^{\mathrm{sub,seed,core}}$*.*

Proof. Item (1) is exactly the combined content of Propositions 1, 2, and 3. Those propositions show that no seed-faithful presentation may vary the benchmark-relevant seed core while still inducing the same fixed root package.

For item (2), Proposition 1 gives the canonical identifications on the seven minimizer images, while Proposition 2 gives the canonical identifications on the shell, support shell, and zero core. Because each of those loci projects to the same fixed lower datum, the seed-faithful equivalence is unique.

For item (3), Proposition 2 shows that the shell chart, support recorder, support-shell chart, zero chart, split, and equation-side transport are carried identically under the canonical seed-faithful equivalence. Proposition 3 then shows that the hidden charge, exact seed zero-response realization, solvability statement, and coercive control are likewise transported identically. Hence the entire benchmark-relevant seed core agrees under the unique seed-faithful equivalence, and it induces the same projected root package.

Item (4) is the project-level consequence of the previous items. Any remaining off-shell freedom outside the benchmark-relevant seed core is not benchmark relevant on the current line, because it does not change the fixed root package or the benchmark outputs. Therefore the current honest root burden has been removed in the only sense that matters for the active one-metric derivational program. □

Corollary 1 (No remaining benchmark-relevant root choice)**.** *At current scope there is no second benchmark-relevant root presentation of the recorded root package other than the projected image of a seed-faithful relabeling of* $\mathfrak{S}_\sigma^{\mathrm{sub,seed,core}}$ *in each sector.*

Proof. If a second benchmark-relevant root presentation existed, it would either change some constituent proved necessary in Theorem 1, contradicting item (1), or else arise from a seed-faithful

relabeling of the same forced seed core, in which case it would not define a genuinely different benchmark-facing root presentation. □

7. Routing Consequence

Corollary 2 (What must be done next)**.** *After Theorem 1, the next honest primary manuscript below the current frontier must do at least one of the following:*

1. *derive or justify the root-generating substrate seed dynamics from still more primitive explicit substrate structure in a way that changes benchmark-facing status;*
2. *identify a genuinely smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside the recorded seed package* $\mathfrak{S}_{\sigma}^{\mathrm{sub,seed}}$*;*
3. *do either of the previous two while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary.*

It is no longer enough merely to restate the current root package $\mathfrak{R}^{\mathrm{sub,rt}}$*, the current foundation package* $\mathfrak{F}^{\mathrm{sub,fdn}}$*, the current ground package* $\mathfrak{G}^{\mathrm{sub,grd}}$*, the current origin package* $\mathfrak{O}^{\mathrm{sub,orig}}$*, the current precursor package* $\mathfrak{P}^{\mathrm{sub,prec}}$*, the current lift package* $\mathfrak{L}^{\mathrm{sub,lift}}$*, the already forced shell-restriction package* $\mathfrak{S}^{\mathrm{sub,sh}}$*, the already forced split selector-shell core* $\mathfrak{C}^{\mathrm{sel,split}}$*, the already forced selector law* $\mathfrak{S}^{\mathrm{sel},0}$*, the already forced combined response law* $\mathfrak{R}^{\mathrm{comb}}$*, the already forced ambient package* $\mathfrak{A}^{\mathrm{bal}}$*, the already forced trace core* $\mathfrak{C}^{\mathrm{tr}}$*, the already forced completion-core presentation* $\mathfrak{C}^{\mathrm{zr,aff}}$*, the already forced exact-shell affine nucleus* $\mathfrak{N}^{\mathrm{aff}}$*, or to insert another same-output relay above the fixed seed core.*

Proof. Theorem 1 removes the benchmark-relevant freedom presently visible at root level. The next honest burden therefore lies either below that level or in a strictly smaller internal compression of it. Another package-preserving restatement of the same root package or another same-output relay above the fixed seed core would not change project status. □

8. Conclusion

The positive result of this note is not that a final substrate microphysics has already been found. Its gain is narrower and more honest. The current foundation-forcing theorem had already shown that the foundation package is the canonical projected exact-shell output of a deeper root class. The present theorem then removes the remaining benchmark-relevant freedom inside the root-facing burden itself. At current scope, the benchmark-relevant seed core that yields the recorded root package is necessary and unique up to canonical seed-faithful equivalence.

That conclusion is exactly the sharper theorem-level gain the project needed below foundation scope. The seed-to-root transport, the seed minimizer ladder, the exact seed shell/support/zero transport, the charted zero/hidden split, the hidden charge, the exact seed zero-response realization, and the solvability / coercive package are no longer merely one possible deeper presentation of the same downstream line. Once the presentation is required to induce the fixed root package, those constituents are forced.

The scope remains disciplined. The benchmark package is unchanged: the one operative metric is still exactly $g^{\sharp}$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. This remains a theorem inside the active derivational

program of Æther-flow Interpretation of Relativity, not a basis for stronger promotion language. The next honest burden is deeper again: derive or justify the root-generating substrate seed dynamics from still more primitive explicit substrate structure, or else prove a genuinely smaller collapse inside the current seed package. Until then, adoption and derivation should continue to be kept sharply separated.